Oil Spill Detection Based on Deep Convolutional Neural Networks Using Polarimetric Scattering Information From Sentinel-1 SAR Images

Xiaoshuang Ma, *Member, IEEE*, Jiangong Xu, Penghai Wu, *Member, IEEE*, and Peng Kong

***Abstract*— Oil spill accidents can cause severe ecological disasters; hence, the timely and effective detection of oil spills on the marine surface is of great significance. Synthetic aperture radar (SAR) is very suitable for large-scale oil spill monitoring. As a more advanced form of SAR, polarimetric SAR (PolSAR) can provide more scattering information of land objects, which can help to improve the accuracy of oil spill detection. However, the current studies of oil spill detection by SAR data have mainly focused on using SAR intensity or amplitude information, and the phase information and other polarimetric information have not been fully utilized. To solve this problem, using Sentinel-1 dual-polarimetric images as the data source, this article presents an intelligent oil spill detection architecture based on a deep convolutional neural network (DCNN), in which both the amplitude information and phase information are utilized. Furthermore, to improve the feature discrimination capability, the Cloude polarimetric decomposition parameters are also integrated into the proposed model. The results show that the improved DeepLabv3+ model, which takes ResNet-101 as the backbone network and group normalization (GN) as the normalization layer, can achieve superior performance than those traditional methods. Moreover, the model is better able to capture the fine details of oil spill instances and can achieve fine-scale segmentation.**

***Index Terms*— Deep convolutional neural network (DCNN), oil spill detection, polarimetric decomposition, polarimetric synthetic aperture radar (PolSAR) image, Sentinel-1.**

I. Introduction

With the increasing pressure on land-based energy sources and the global economy's continuous development, offshore oil transport has become an important way for people to obtain energy [1]. At the same time, oil spills occur from time to time, and oil pollution in the sea is becoming increasingly severe [2], [3]. According to the report of the International Tanker Owners Pollution Federation (ITOPF), over the past 50 years, oil tankers have spilled approximately 5.86 million tons of oil due to accidents, which has caused severe harm to ocean ecological security [4]. Therefore, accurate detection of oil spills is crucial for conducting a clean-up operation and reducing the ecological risk.

Remote sensing now plays a vital role in oil spill detection [5], [6]. Synthetic aperture radar (SAR) can actively transmit signals to detect land targets, working all day and in all weather conditions. Normally, the oil film can attenuate the Bragg waves on the ocean surface, thereby forming a dark spot that differs from the brightness of the other surrounding areas in SAR images [7], [8]. However, other phenomena, such as low wind speed regions, algal blooms, and wave shadows behind land, can also have low backscattering characteristics [9], [10], which leads to difficulty in distinguishing oil spills from look-alikes. As an advanced imaging radar technology, polarimetric SAR (PolSAR) systems can obtain different polarization information of the target and fully reveal its physical scattering mechanism [11]. PolSAR is, therefore, effective in enhancing the distinction between oil spills and other look-alike occurrences with low backscattering characteristics [12]. As a result, PolSAR is now one of the primary means of oil spill detection on the marine surface.

Researchers have conducted in-depth studies using polarimetric information to identify marine oil spills in PolSAR images [16]–[20]. For example, Migliaccio *et al.* [13] proposed to detect dark spots in PolSAR images based on the constant false alarm rate (CFAR) filter and used polarimetric target decomposition theory to distinguish oil spills and other similar phenomena. Subsequently, Nunziata *et al.* [14] used multilook complex (MLC) C-band SAR images to demonstrate that the copolarized phase difference (CPD) can be used to distinguish oil spills and biogenic slicks. Skrunes *et al.* [15] used Radarsat-2 images and TerraSAR-X images to compare the ability of multipolarization features, such as the entropy (H), mean scattering angle (α), and standard deviation of the CPD in detecting different oil types. In [16], the validity of the degree of polarization (DoP) for oil spill detection under different polarimetric situations and incidence angles in dual-PolSAR imagery has been demonstrated. To reveal the importance of selecting the optimal polarimetric features for oil spill identification, Xu *et al.* [17] compared the identification performances using several different polarimetric parameters and classification methods. Although researchers

Manuscript received June 5, 2021; revised September 15, 2021 and October 31, 2021; accepted November 1, 2021. Date of publication November 8, 2021; date of current version February 15, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 41701390 and in part by the Hefei Municipal Natural Science Foundation under Grant 2021041. *(Corresponding author: Xiaoshuang Ma.)*

Xiaoshuang Ma, Jiangong Xu, and Penghai Wu are with the School of Resources and Environmental Engineering, the Anhui Province Key Laboratory of Wetland Ecosystem Protection and Restoration, and the Information Materials and Intelligent Sensing Laboratory of Anhui Province, Anhui University, Hefei 230601, China (e-mail: mxs.88@whu.edu.cn; dd_xjg@163.com; wuph@ahu.edu.cn).

Peng Kong is with the Institute of Spacecraft System Engineering, Beijing 100094, China (e-mail: kpnkpn@126.com).

Digital Object Identifier 10.1109/TGRS.2021.3126175

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:07:39 UTC from IEEE Xplore. Restrictions apply.

have thoroughly analyzed the feasibility of using polarimetric information to detect oil spills, it is still difficult to fully mine the polarimetric information of the image with only a few manually selected parameters. Moreover, satisfactory continuous segmentation results can hardly be obtained by most traditional methods.

The application of deep learning in remote sensing image classification has attracted extensive attention. Deep learning has a strong capability of representation learning and can automatically mine information from massive data to improve classification performance [18], [19]. The deep convolutional neural network (DCNN) is one of the representative algorithms of deep learning. DCNNs focus on the spatial correlation of the data and can significantly reduce the number of variables that need to be learned by using a parameter sharing mechanism [20]. SAR systems have the ability to acquire multitemporal data, and thus, data-driven deep learning is a promising way to make the best use of SAR data [21]. Some studies have also been carried out for oil spill detection from PolSAR images using deep neural networks [6], [25]–[29]. For example, Krestenitis *et al.* [6] published an SAR image dataset for oil spill detection and analyzed the performance of the typical DCNN segmentation models with the dataset. Using quad-PolSAR image, Chen *et al.* [22] applied a stacked autoencoder (SAE) and a deep belief network (DBN) to optimize the polarimetric feature sets for oil detection, which showed better performances than support vector machine (SVM) and the traditional artificial neural networks (ANNs). Zeng and Wang [23] designed a network with 12 weight layers named the oil spill convolutional network (OSCNet), which clearly improved the detection accuracy. Besides, Bianchi *et al.* [24] proposed a deep learning framework to segment and identify oil spills in SAR datasets. In [25] and [26], the researchers proposed to use the single-bounce eigenvalue relative difference (SERD) and Yamaguchi parameters as the input of DCNN models for oil detection.

The aforementioned studies have shown higher accuracy of oil spill detection compared to the traditional machine learning (ML)-based approaches. However, most of these studies have only focused on using the intensity or amplitude information of SAR images, and other polarimetric information, which can better reflect the physical characteristics of the targets, has been ignored. As mentioned before, other look-alikes, such as low wind speed regions, algal blooms, and wave shadows behind land, also show very low intensity or amplitude values as oil spills. As a result, misclassification results between oil spills and look-alikes could be serious for the detection algorithms that only use the intensity information. PolSAR systems obtain the scattering characteristics of the targets in different polarimetric states, and the introduction of more polarimetric information for oil spill detection can help to improve detection accuracy. Furthermore, most of the studies have utilized only a small amount of sample data, resulting in the poor generalization performance of the models.

In this article, we focus on the topic of oil spill detection from Sentinel-1 SAR images by effectively integrating multiple polarimetric features with a DCNN. First, the polarimetric covariance matrix and the polarimetric decomposition parameters from the Sentinel-1 data are employed as the model input. Then, a semantic segmentation network that is suitable for the PyTorch framework is implemented to extract the oil spill information. Based on this classical DCNN structure, the influence of different configurations that take different backbone networks as the deep feature extraction layer and different normalization methods as the regularizer for detecting oil spills was further investigated.

TABLE I
BASIC INFORMATION OF THE DATA SOURCE

Satellite	Sentinel-1
Polarimetric	Dual (VV+VH)
Product Type	SLC
Product Level	L1
Mode	IW
Band	C
Swath	250 km
Spatial resolution	5 m × 20 m
Incidence Angle	33.86–42.93°

The rest of this article is organized as follows. Section II briefly introduces the basic information of Sentinel-1 data and describes the development process of the dataset in detail. Section III mainly describes the DeepLabv3+ architecture used in this research and the improved method and presents the evaluation metrics. In Section IV, we described the experimental results and conducted comparative analyses. Finally, our conclusions are drawn in Section V.

II. DATASET DESCRIPTION

The Sentinel-1 data were downloaded from the European Space Agency (ESA) data distribution website (https://scihub.esa.int). Sentinel-1 is an Earth observation satellite for environmental and security monitoring that is operated under ESA's Copernicus Programme. The Sentinel-1 sensor that is equipped with a C-band imaging system has effectively continued the Earth observation missions of ERS-1/2 and ENVISAT ASAR, employing four operation modes, namely, Stripmap (SM), Interferometric Wide (IW) Swath, Extra-Wide (EW) Swath, and Wave (WV) [27]. As an innovative SAR system, the Sentinel-1 mission comprises a constellation of two polar-orbiting satellites. It not only provides dual-polarimetric capability but also has very short revisit times and rapid product delivery. As a result, Sentinel-1 can provide strong data support for global environmental monitoring. In this research, we used Sentinel-1 single-look complex (SLC) products. The basic information of the data source is provided in Table I.

In this study, we acquired 35 Sentinel-1 images, covering a period from October 4, 2014, to September 25, 2020, with a relatively concentrated distribution in Europe and Asia, especially the Mediterranean Sea, the North Sea, the Red Sea, and the Persian Gulf. Information regarding the location and time of the oil spill events was mainly collected from some official agencies, such as the Ministry of Environment Malaysia and the National Oceanic and Atmospheric Administration (NOAA). Some information about the oil spill events

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:07:39 UTC from IEEE Xplore. Restrictions apply.

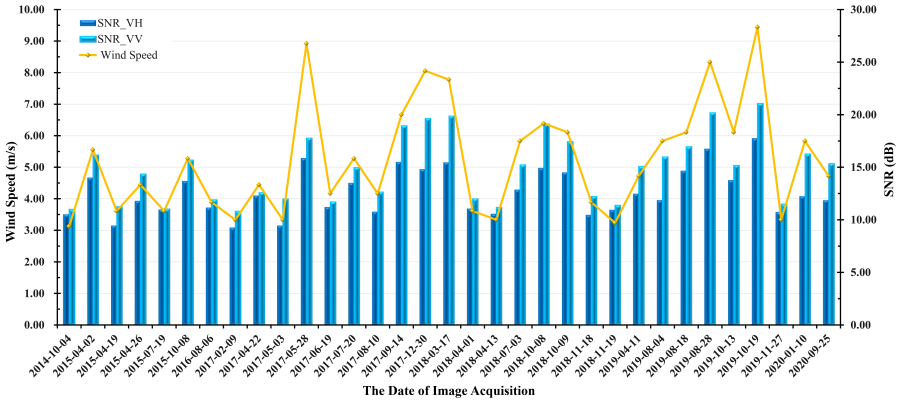


Fig. 1. Wind speed information and SNR values of the data source, where the wind speed information is acquired from https://earth.nullschool.net/zh-cn/.

can be also obtained from relative works [28]–[30]. (The dataset and the source code of this article are available at https://github.com/RS-xjg/oil-spill-detection.)

Besides the inherent scattering traits of the oil spills, the polarimetric characteristics of the oil-covered sea surface also depend on some other factors, such as imaging parameters (e.g., polarization, incidence angle, and noise level) [31], [32] and environmental conditions (e.g., local wind vector) [33], [34]. As revealed in [33] and [35], the ocean backscattering decreases as the incidence angle increases, and the most useful angles for observing oil spills are between 20° and 45°. As shown in Table I, the range of the incidence angle of Sentinel-1 images used in this study is between 33.86° and 42.93°; hence, the datasets used in this study are suitable for oil spill detection. In addition, it should be pointed out that the studies in [36] and [37] show that non-Bragg scattering mechanisms and the thinner films of oil are observable only in the event of moderate to high wind speed (normally, between 3 and 10 m/s); moreover, in the case of low signal-to-noise ratio (SNR) (normally, lower than 10 dB), the H/A/alpha values are sensitive and might not be a consequence of actual volume scattering [38], [39]. The wind speed information and SNR values of the data source used in this study are provided in Fig. 1. As we can see, the wind speed values are generally between 3 and 10 m/s, which is suitable for oil detection by SAR satellites; besides, the SNR values are all higher than 10 dB, indicating that the H/A/alpha decomposition parameters are suitable to distinguish oil spills and sea surface.

A. Preprocessing Scheme

To retain the PolSAR image information to the maximum extent, we only conducted simple, yet necessary, data processing. The preprocessing scheme involves the following steps.

1) An area containing the oil spill area and other relevant information (sea surface, land, and so on) was cropped from the original PolSAR).
2) We then performed radiometric calibration on the images to transform the original images into quantity values with practical physical significance to reduce the influence of radiation deviation on the backward coefficients of the radar with different surfaces.
3) SAR images exhibit strong speckle noise due to the coherent imaging mechanism, which seriously affects the visual interpretation and information extraction. In this study, a 7×7 refined Lee filtering method was used to perform filtering and smoothing of the homogeneous regions.

B. Polarimetric Covariance Matrix

Each pixel of a PolSAR image contains both amplitude and phase information, which can be represented by a 2×2 complex scattering matrix [40]

$$S = \begin{bmatrix} S_{\mathrm{HH}} & S_{\mathrm{HV}} \\ S_{\mathrm{VH}} & S_{\mathrm{VV}} \end{bmatrix} \tag{1}$$

where the subscript of each element represents the polarimetric state of the SAR antenna, such that S_{HV} is the scattering coefficient between the vertical transmitting and horizontal receiving polarizations. Under a monostatic case, according to the reciprocity theorem, the scattering matrix S is symmetrical, namely, $S_{\mathrm{VH}} = S_{\mathrm{HV}}$.

Sentinel-1 data used in this article are VV-VH dual-polarization images, and its scattering matrix is defined as

$$S = \begin{bmatrix} 0 & 0 \\ S_{\mathrm{VH}} & S_{\mathrm{VV}} \end{bmatrix}. \tag{2}$$

Then, the target vector [41] is defined as

$$K = [S_{\mathrm{VH}} S_{\mathrm{VV}}]^T \tag{3}$$

where T denotes the conjugate transpose.

The polarimetric covariance matrix of a pixel in the dual-polarimetric image can be defined as:

$$C = \langle K K^{*T} \rangle = \begin{bmatrix} \langle 2|S_{\mathrm{VH}}|^2 \rangle & \left\langle \sqrt{2} S_{\mathrm{VH}} S_{\mathrm{VV}}^* \right\rangle \\ \left\langle \sqrt{2} S_{\mathrm{VV}} S_{\mathrm{VH}}^* \right\rangle & \langle |S_{\mathrm{VV}}|^2 \rangle \end{bmatrix} \tag{4}$$

with

$$S_{\mathrm{VH}} = |S_{\mathrm{VH}}| e^{j \emptyset_{\mathrm{VH}}} \tag{5}$$

where $|S_{\mathrm{VH}}|$ and $\emptyset_{\mathrm{VH}}$ represent the amplitude and the phase of the polarization channel, respectively. $*$ is the conjugate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:07:39 UTC from IEEE Xplore. Restrictions apply.

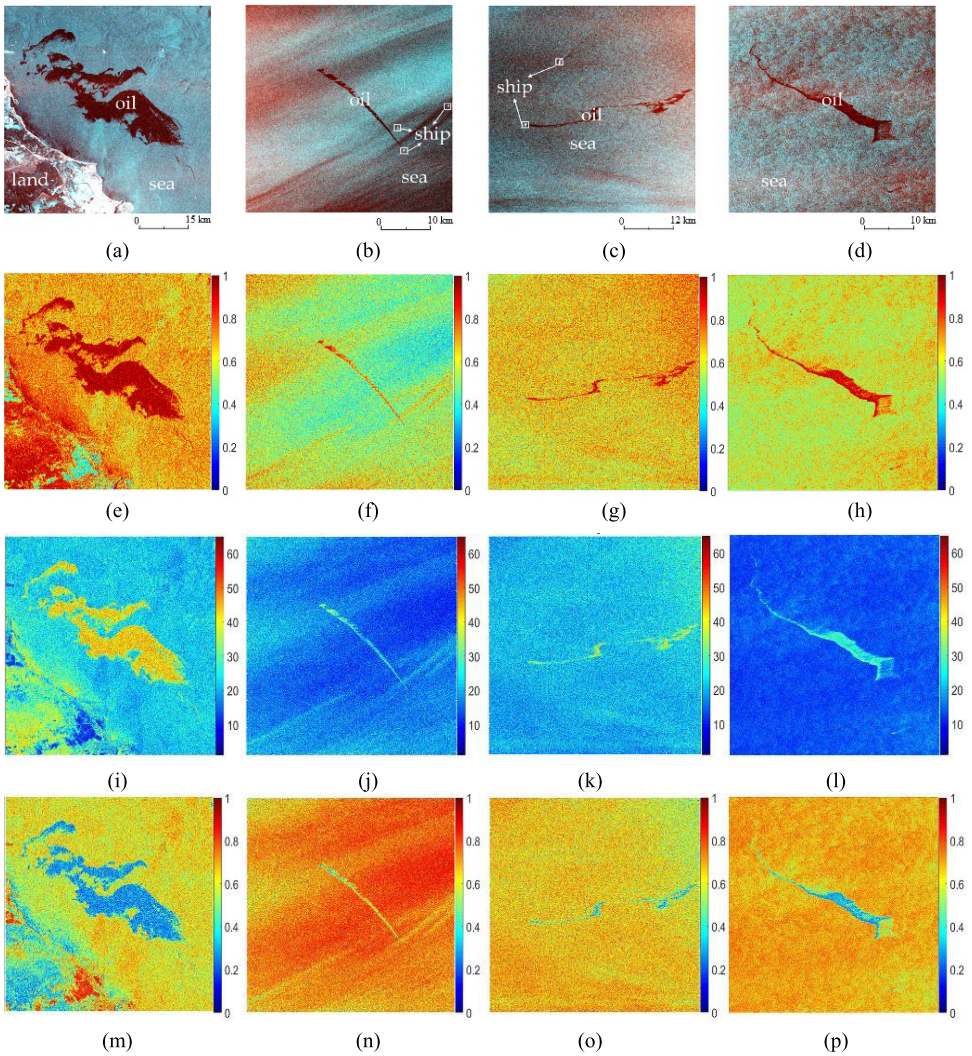


Fig. 2. (a)–(d) Original PolSAR images contain the oil spill areas and other relevant information (sea surface, land, and so on). (e)–(p) H/A/alpha decomposition results for the oil film on the sea surface: (e)–(h) results of H, (i)–(l) results of α, and (m)–(p) results of A.

operator, $\langle\cdot\rangle$ represents the statistical average, and j is the imaginary unit.

C. H/A/Alpha Decomposition

The polarimetric decomposition parameters can reflect the polarimetric scattering characteristics of land objects. In order to make full use of the polarimetric information of the PolSAR images in oil spill detection, we extracted the polarimetric characteristic parameters from the Sentinel-1 dual-polarimetric data by adopting H/A/alpha decomposition [41]. The polarimetric entropy (H), mean scattering angle (α), and polarimetric anisotropy (A) can be obtained by eigenvalue decomposition.

The H/A/alpha decomposition method is originally proposed for fully PolSAR images; it can be also applied for dual-polarimetric data [42]. As we know, the covariance matrix C is the positive semidefinite matrix [43]. Thus, the covariance matrix C can be decomposed into the weighted sum of two mutually orthogonal covariance matrices, expressed as follows:

$$C = \sum_{\tau=1}^{2} \lambda_\tau C_\tau = \lambda_1 e_1 e_1^* + \lambda_2 e_2 e_2^* \tag{6}$$

where λ_τ and e_τ represent the real eigenvalue and the eigenvector of the covariance matrix C, respectively. τ is the rank of the matrix.

In the H/A/alpha decomposition model, H is used to describe the degree of statistical disorder of each distinct scatter type within the ensemble (i.e., the randomness of the scattering), which is defined as

$$H = -\sum_{\tau=1}^{2} p_\tau \log_2 p_\tau, \quad H \in [0, 1] \tag{7}$$

with

$$p_\tau = \frac{\lambda_1}{\lambda_1 + \lambda_2}. \tag{8}$$

A larger H value represents a more chaotic scattering trait.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:07:39 UTC from IEEE Xplore. Restrictions apply.

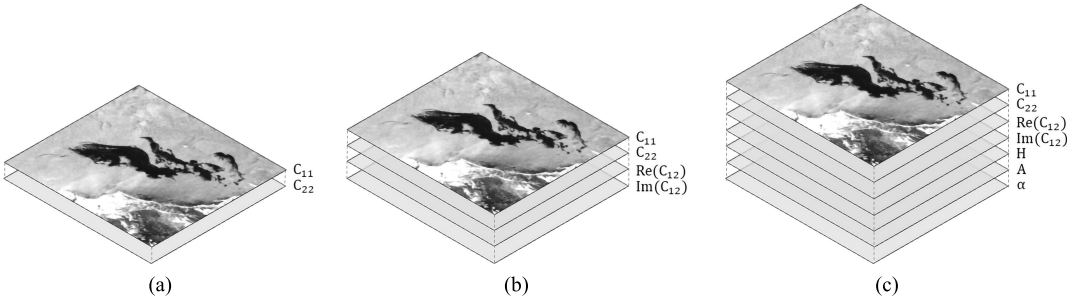


Fig. 3. PolSAR dataset form: (a)–(c) dataset D1, dataset D2, and dataset D3, respectively. Considering the reciprocity theorem and the avoidance of data redundancy, the PolSAR data are constructed into two-, four-, and seven-channel datasets, respectively. C_{11} and C_{22} are the diagonal elements of the polarimetric covariance in (4); Re and Im denote the real and imaginary parts of C_{12}; H, A, and α represent the polarimetric entropy, polarimetric anisotropy, and mean scattering angle, respectively.

For random media, α is the main parameter used for identifying the dominant scattering mechanism (i.e., it is a roll-invariant parameter), as follows:

$$\alpha = \sum_{\tau=1}^{2} p_\tau \cos^{-1}(|e_\tau(1)|), \quad \alpha \in [0°, 90°]. \tag{9}$$

An α value lower than 45° denotes that the radar signal of a rough surface is dominated by the odd scattering process, and with the increase in the α value, the scattering process becomes dominated by double scattering.

A measures the relative importance of the first and second eigenvalues resulting from the eigenvalue decomposition [44], [45], as follows:

$$A = \frac{(\lambda_1 - \lambda_2)}{(\lambda_1 + \lambda_2)}, \quad A \in [0, 1]. \tag{10}$$

When the value of A is low, it shows that there is only one primary type of scattering in the region.

As shown in Fig. 2, the oil areas can be clearly distinguished from the surrounding land objects in the maps of the three polarimetric decomposition parameters. Meanwhile, this also shows that these three parameters can effectively describe the physical scattering characteristics of oil film on the sea surface. Therefore, using H/A/alpha decomposition parameters can help the deep learning method to identify oil spills accurately.

D. Dataset Form

In this study, to thoroughly compare the capabilities of different information of PolSAR images in oil spill detection, we built three types of datasets, as shown in Fig. 3. Among the datasets, dataset D_1 only considers the scattering intensity information of the land objects; dataset D_2 includes the scattering intensity information and the phase information of the land objects; and based on D_2, the H/A/alpha polarimetric decomposition parameters of the land objects is introduced in dataset D_3.

We manually annotated the preprocessed images to obtain the oil spill labels by the LabelMe software, a deep learning annotation tool (https://github.com/CSAILVision/LabelMeAnnotationTool). The pixel values of "0" and "1" in the single-channel labels represent "nonoil spill" and "oil spill," respectively. The total numbers of pixels in categories "0" and "1" are 893, 512, 582, and 50, 205, 818, respectively. Thus, the pixels of class "oil spill" are 5.32% of the total.

TABLE II
NUMBER OF SCENES FOR EACH INSTANCE

Scene	Number
Sea only	725
Land only	191
Oil spill only	15
Look-alikes only	6
Sea + Land	287
Sea + Oil spill	475
Sea + Look-alikes	598
Sea + Land + Oil spill	372
Sea + Land + Look-alikes	251
Sea + Oil spill + Look-alikes	677
Sea + Land + Oil spill + Look-alikes	3
Total	3600

In order to be more suitable for the network input, we cut the image and its labels into subimages of 512 × 512 in size. Therefore, we had 3600 samples. We also enhanced the dataset with horizontal and vertical mirrorings to prevent model overfitting. Finally, the number of samples was 10 800. Then, we randomly split the sample into three independent parts, namely, the training set, the verification set, and the test set, in which the ratio of them was 5:2:3. Table II provides the statistics of different scenes of the dataset. Generally speaking, for most scenes, the number of samples used in this study is sufficient, which can ensure the good generalization capability of the proposed model.

III. EXPERIMENTAL METHOD

A. Basic Architecture of DeepLabv3+

The fully supervised learning methods are significantly more effective than the weakly supervised and semisupervised learning methods [46]. The encoder–decoder structure is a popular network structure in semantic segmentation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:07:39 UTC from IEEE Xplore. Restrictions apply.

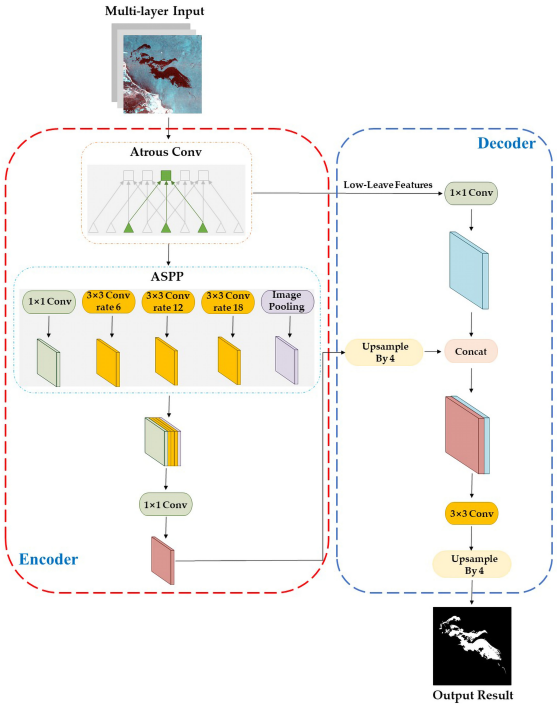


Fig. 4. Encoder–decoder architecture for oil spill detection.

The encoder obtains the pixel position information by convolution, and the pooling operation decoder restores the pixel position information by deconvolution, thus forming a symmetric network structure. DeepLabv3+ [47] relies on the DeepLabv3 [48] framework by adding a decoder to refine the segmentation results, making it one of the most effective semantic segmentation algorithms currently available. DeepLabv3+ has achieved satisfactory results in many public datasets, such as the PASCAL VOC2012 dataset [49] and the CityScapes dataset [50]. The structure of DeepLabv3+ is shown in Fig. 4. It can be seen that the network consists of an encoder and a decoder. The encoder utilizes the atrous spatial pyramid pooling (ASPP) [51] module to fully mine the image's multiscale context information. The decoder uses the shallow feature map to optimize the location information that cannot be recovered by upsampling to obtain the semantic segmentation prediction results. In this study, an improved DeepLabv3+ model was used for the oil spill detection in PolSAR images. The full details of the DeepLabv3+ network are provided in the following.

DeepLabv3+ adds atrous convolution to the deep feature extraction network. Based on reducing the number of downsampling operations and not increasing the network parameters, the network's receptive field is increased. Making sure that the feature map does not lose the boundary feature information of the segmentation target, as much as possible, can improve the segmentation effect [52].

The DeepLabv3+ model adopts the spatial pyramid pooling (SSP) operation in SSP-Net [53] and improves it into ASPP to achieve the segmentation capability for multiscale targets. ASPP performs 1×1 convolution, 3×3 convolution with expansion rates of 6, 12, and 18, and global average pooling of the input feature maps. The feature maps are then

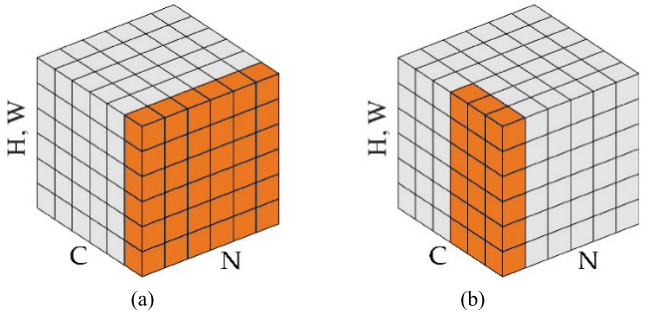


Fig. 5. 3-D display of (a) BN and (b) GN, where N is the batch axis, C is the channel axis, and (H, W) represents the spatial axis. The pixels in orange are normalized by the same mean and variance.

fused, and 1×1 convolution is performed to compress the number of channels to 256. Accordingly, ASPP can extract and distinguish the feature information of targets at different scales and achieve the segmentation of multiscale targets.

Finally, DeepLabv3+ performs the necessary upsampling operation on the input image to fully extract the high-level feature information of the target object in the image. To compensate for the boundary information lost in the downsampling operation, DeepLabv3+ adopts an encoder–decoder structure to fuse the low-level features in the feature map recovery process and recover some of the boundary information of the target. The feature map recovery adopts a linear interpolation method, which improves the accuracy of the network segmentation.

B. Normalization Layer

The normalization layer plays an important role in convolutional neural networks by preventing gradient explosion and gradient disappearance. The semantic segmentation algorithms based on a batch normalization (BN) layer can achieve high accuracy in image segmentation, but the use of the BN layer is affected by the batch size (BS). When the BS is too small, the model's accuracy is significantly reduced, and when the BS is too large, the usage of memory resources is increased. Therefore, the BS and computer memory capacity should be considered when training a model based on BN to achieve the best training effect. According to the problems encountered in BN, group normalization (GN) [54] was adopted as an alternative to BN. The GN is used as the normalization layer, in which the channels are divided into groups, and the features are normalized within each group to calculate the mean value and variance of each group of channels. The most significant advantage of GN is that it is not affected by the BS. The BN and GN layers are also 3-D, as shown in Fig. 5. Therefore, a GN layer is adopted in the DCNN model proposed in this article, which is expected to further improve the accuracy of oil spill detection without occupying too much memory.

The calculation formulas for the mean value (μ) and standard deviation (σ) are given as follows:

$$\mu_i = \frac{1}{m} \sum_{k \in S_i} x_k \tag{11}$$

$$\sigma_i = \sqrt{\frac{1}{m} \sum\nolimits_{k \in S_i} (x_k - u_i)^2 + \epsilon} \tag{12}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:07:39 UTC from IEEE Xplore. Restrictions apply.

where m represents the size of this set and S_i is the set of pixels in which the mean and standard deviation are computed. ϵ is a small constant, normally set as 1×10^{-5}.

In the BN layer [55], the set S_i is given as follows:

$$S_i = \{k | k_C = i_C\} \tag{13}$$

where i_C (and k_C) denotes the subindex of i (and k) along the channel axis.

In GN, the set S_i is given as follows:

$$S_i = \left\{ k \mid k_N = i_N, \left\lfloor \frac{k_C}{C/G} \right\rfloor = \left\lfloor \frac{i_C}{C/G} \right\rfloor \right\} \tag{14}$$

where G is the number of groups and C/G is the number of channels per group.

C. Experimental Details

In pixel-level image classification, the scale of the dataset affects the training effect of the model. Therefore, we used geometric methods, such as vertical mirroring and horizontal mirroring, to perform image augmentation operations, which can expand the dataset to improve the fitting effect of the model.

Based on the oil spill detection datasets (D_1, D_2, and D_3) constructed in this study, we used the DeepLabv3+ network to carry out semantic segmentation training, with Xception [56], ResNet-101 [57], and MobileNetv2 [58] used as backbone networks for comparison. Considering that BN in the training model has some limitations, we adopted GN as the preferred solution. On the basis of setting the initial hyperparameters of the model, we adjust the hyperparameters to achieve optimization of the DeepLabv3+ network.

In addition, to fully demonstrate the superiority of the improved DeepLabv3+ model in detecting oil spills from dual-PolSAR images, two traditional ML methods were also considered, i.e., Wishart supervised classification and SVM, in addition to three typical semantic segmentation networks, i.e., a fully convolutional network (FCN) [59], SegNet [60], and U-Net [61]. These methods were experimentally compared with the improved DeepLabv3+ model in terms of both accuracy and processing time.

In this study, a series of well-known semantic segmentation networks suitable for the PyTorch framework were used to extract oil spill information from Sentinel-1 dual-polarimetric images. The hardware and software configurations, including the framework version and calculation platform, are shown in Table III.

Due to category imbalance in the dataset, direct training may lead to model nonconvergence or overfitting phenomena. Therefore, we adopted focal loss (FL) [62], which modifies the weight coefficient to reduce the weight occupied by negative samples in the training and to mine complex samples, as follows:

$$\mathrm{FL}(p_t) = -\beta_t (1 - p_t)^{\gamma} \log(p_t) \tag{15}$$

where γ is a tunable focusing parameter that reduces the weight of the easy classification samples to make the model focus more on the difficult classification samples; β is a weighting factor, normally between 0 and 1; and t is the current classification. According to the study in [62] and by a number of experiments, we found that, when γ was set as 2 and β was set as 0.25, the training results can approach optimal.

TABLE III
HARDWARE AND SOFTWARE CONFIGURATIONS OF THE EXPERIMENTS

Configuration	Version
GPU	NVIDIA GeForce GTX 1070 Ti
Memory	16.0 GB
System	Windows 10 Enterprise, 64-bit
Language	Python 3.8.3
Programming	PyCharm 2020.2.1
Frame	PyTorch 1.6.0
CUDA	11.1.96
CUDNN	7.6.4

In order to achieve the optimal configuration of hyperparameters, we carried out a large number of adjustment experiments for the setting of the different hyperparameters. The hyperparameters that we chose to adjust were the learning rate (LR) of the Adam optimizer $\{1 \times 10^{-5}, 1 \times 10^{-4}, 1 \times 10^{-3}\}$, the BS $\{2, 4, 6, 8\}$, and the normalization layer {BN, GN}. Through the experimental testing, we found that the use of GN for the normalization layer, LR $= 1 \times 10^{-3}$, and BS $= 2$ was the optimal configuration.

In the model training, the LR was automatically adjusted with the increase in the number of training epochs, i.e., the LR decreased by ten times every 15 epochs. In addition, to accelerate the convergence speed of the network, we introduced the Adam optimizer to optimize the network and update the parameters. In this study, we adopted the holdout validation training method [63] for the models, i.e., the training set and the validation set were input into the models simultaneously. The final predictive model is then evaluated with the test set.

D. Evaluation Metrics

In order to objectively test the performance of the different models, we calculated a series of evaluation indices, i.e., the overall accuracy (OA), $F1$-score ($F1$), and intersection over union (IoU).

OA represents the proportion of the correctly classified pixels to total pixels and is defined as

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{16}$$

where TP, TN, FP, and FN denote the number of true positives, true negatives, false positives, and false negative samples, respectively.

$F1$ can be regarded as a harmonic average of the model precision and recall rate, which is defined as

$$F1 = \frac{2 * \mathrm{TP}}{2 * \mathrm{TP} + \mathrm{FN} + \mathrm{FP}}. \tag{17}$$

IoU is the ratio of the intersection and union of the prediction and the ground truth. IoU_0 and IoU_1 represent the IoU of "nonoil spill" and "oil spill," respectively. The mean IoU

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:07:39 UTC from IEEE Xplore. Restrictions apply.

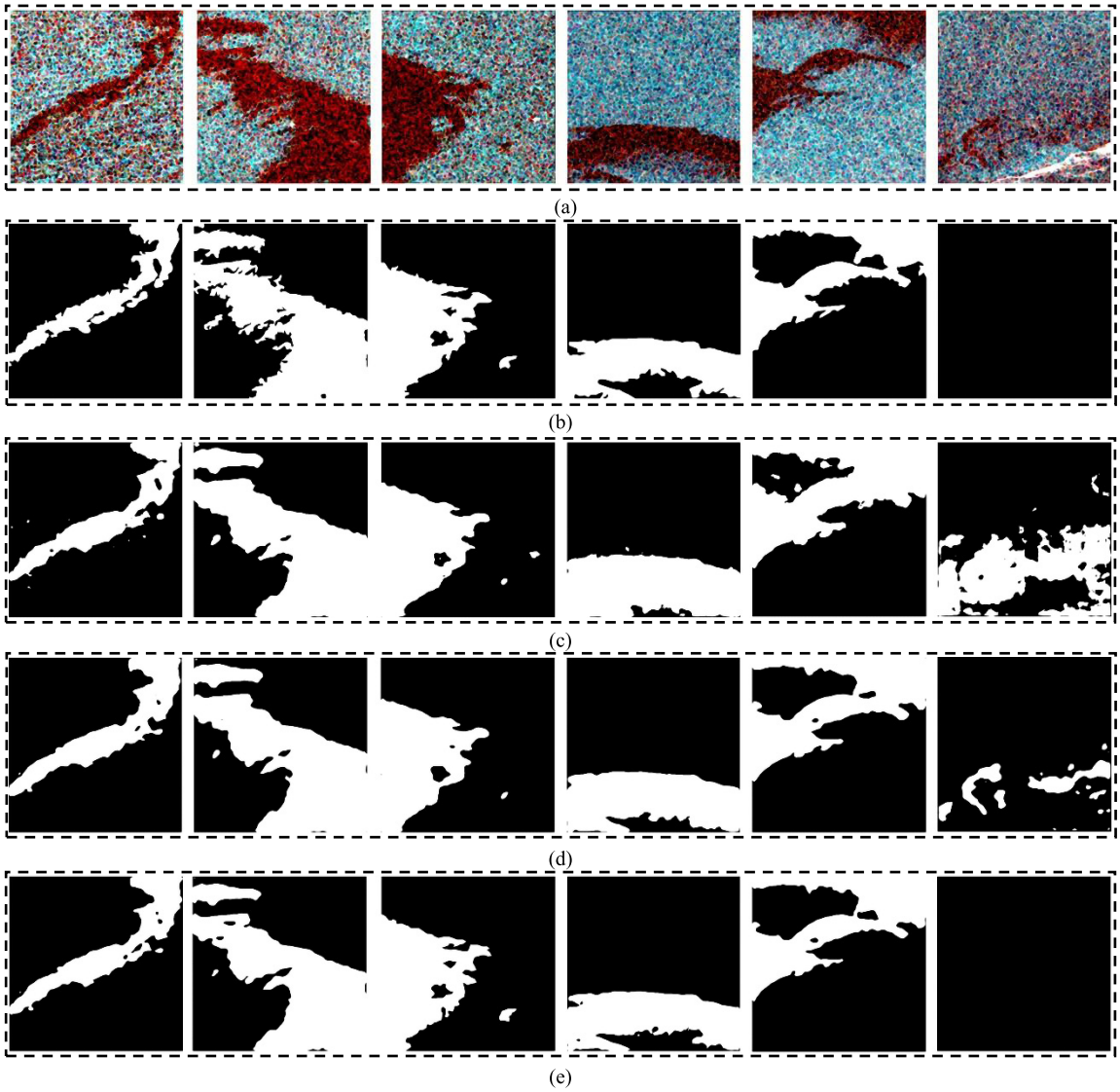


Fig. 6. (a) Original SAR images. (b) Ground truth of the SAR images. Examples of the detection results of on (c) dataset D_1 (only considers the scattering intensity information of the land objects), (d) dataset D_2 (includes the scattering intensity information and the phase information of the land objects), and (e) dataset D_3 (based on D_2, the $H/A/$alpha polarimetric decomposition parameters of the land objects are introduced) by using the DeepLabv3+ model, in which the backbone network is ResNet-101, and the normalization layer is GN.

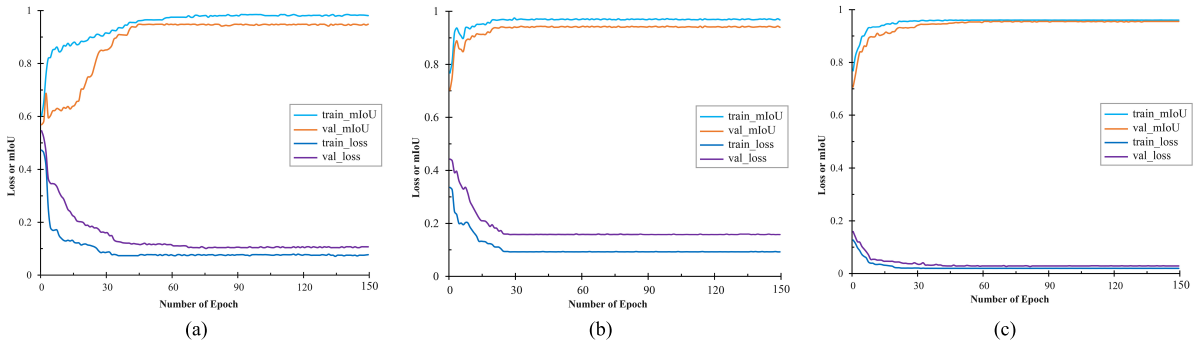


Fig. 7. (a)–(c) mIoU and loss curves of DeepLabv3+ on dataset D_3 for MobileNetv2, Xception, and Resnet-101 backbones, respectively.

(mIoU) is the average IoU measured over each class. These indices can be defined as

$$\text{IoU_0} = \frac{\text{TN}}{\text{FN} + \text{TN} + \text{FP}} \tag{18}$$

$$\text{IoU_1} = \frac{\text{TP}}{\text{FP} + \text{TP} + \text{FN}} \tag{19}$$

$$\text{mIoU} = \frac{1}{2}\left(\frac{\text{TP}}{\text{FP} + \text{TP} + \text{FN}} + \frac{\text{TN}}{\text{FN} + \text{TN} + \text{FP}}\right). \tag{20}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:07:39 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
PRECISION OF DEEPLABV3+ UNDER DIFFERENT SETTINGS (%)

Dataset	Backbone	Normalization	OA	F1	IoU_0	IoU_1	mIoU
D_1	Xception	BN	84.73	76.40	94.38	59.48	76.93
	MobileNetv2		82.42	73.06	93.27	52.17	72.72
	ResNet-101		83.22	74.55	94.7	55.35	75.03
	Xception	GN	88.89	77.16	93.19	58.19	75.69
	MobileNetv2		87.38	76.30	95.62	57.45	76.54
	ResNet-101		88.41	79.57	95.84	58.6	77.22
D_2	Xception	BN	89.32	81.03	98.26	61.81	80.04
	MobileNetv2		82.12	87.89	96.1	76.84	86.47
	ResNet-101		90.19	87.43	97.42	81.66	89.54
	Xception	GN	91.79	77.86	98.39	63.75	81.07
	MobileNetv2		91.43	89.76	98.22	76.62	87.42
	ResNet-101		92.79	90.04	98.38	82.5	90.44
D_3	Xception	BN	92.40	88.42	98.83	85.82	92.33
	MobileNetv2		96.37	92.33	99.45	87.76	93.60
	ResNet-101		97.35	92.67	99.47	88.35	93.91
	Xception	GN	93.19	92.76	**99.72**	89.19	94.46
	MobileNetv2		97.88	93.83	99.11	90.83	94.97
	ResNet-101		**98.92**	**94.38**	99.52	**91.58**	**95.55**

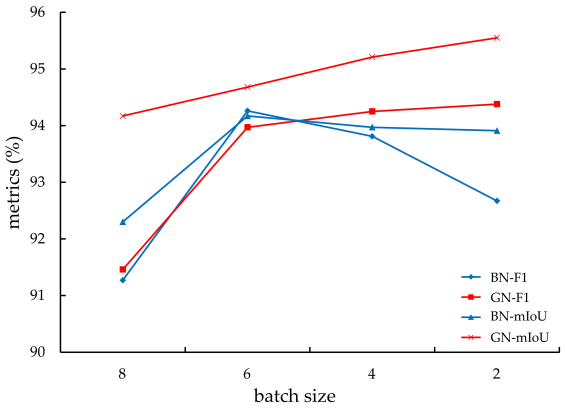


Fig. 8. Line chart of different normalization layers.

IV. EXPERIMENTAL RESULTS

In this section, we analyze the capability of each model with the built dataset for oil spill detection from two aspects, i.e., accuracy and processing time. In particular, we describe the detection effect of the improved DeepLabv3+ model.

A. Comparison of DeepLabv3+ With Different Setups

We comprehensively and accurately evaluated the performance of DeepLabv3+ from the three aspects of the model input, the backbone network, and the normalization method. The precision of the DeepLabv3+ model under different settings is reported in Table IV (the best results are in bold). Overall, when the network input is dataset D_3, the precision of the model is clearly better than with the other datasets (i.e., dataset D_1 and dataset D_2). Xception and MobileNetv2 have comparable performances in the backbone network setup, but ResNet-101 obtains a slightly higher accuracy. The different settings of the normalization layer significantly influence the precision of the network, and the precision of GN is significantly higher than that of BN.

Fig. 6 displays the detection results of the DeepLabv3+ model, in which the backbone network is ResNet-101 and the normalization layer is GN using Dataset D_1, Dataset D_2, and Dataset D_3, respectively. From Fig. 6, it can be seen that, when the model input is dataset D_3, i.e., the amplitude information, phase information, and polarimetric decomposition parameters, the classification results are close to the actual labels. Table IV and Fig. 6 indicate that the detection method based on multifeature fusion has significant advantages. As the sea surface environment is complex, ships and look-alikes may affect the detection effect for oil spills. As shown in Fig. 6, there are look-alikes in the original image. However, according to the detection results, when the model inputs are dataset D_1 and dataset D_2, the look-alikes are more likely to be misclassified as oil spills. There is also the problem of incomplete extraction of oil spill areas. The above issues imply that using PolSAR images to distinguish oil spills and look-alikes is challenging. As we mentioned previously, taking account of multiple polarimetric information can help to improve the oil spill detection performance.

To improve the performance of the DeepLabv3+ network for oil spill detection, Xception, Resnet-101, and MobileNetv2 backbones were selected and compared in this study. Fig. 7 shows the mIoU and loss curves of the oil spill detection by DeepLabv3+ using the different backbone networks. When the backbone networks are set as Xception and MobileNetv2, the curve fluctuates considerably, and there are apparent differences between the training process and the test process,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:07:39 UTC from IEEE Xplore. Restrictions apply.

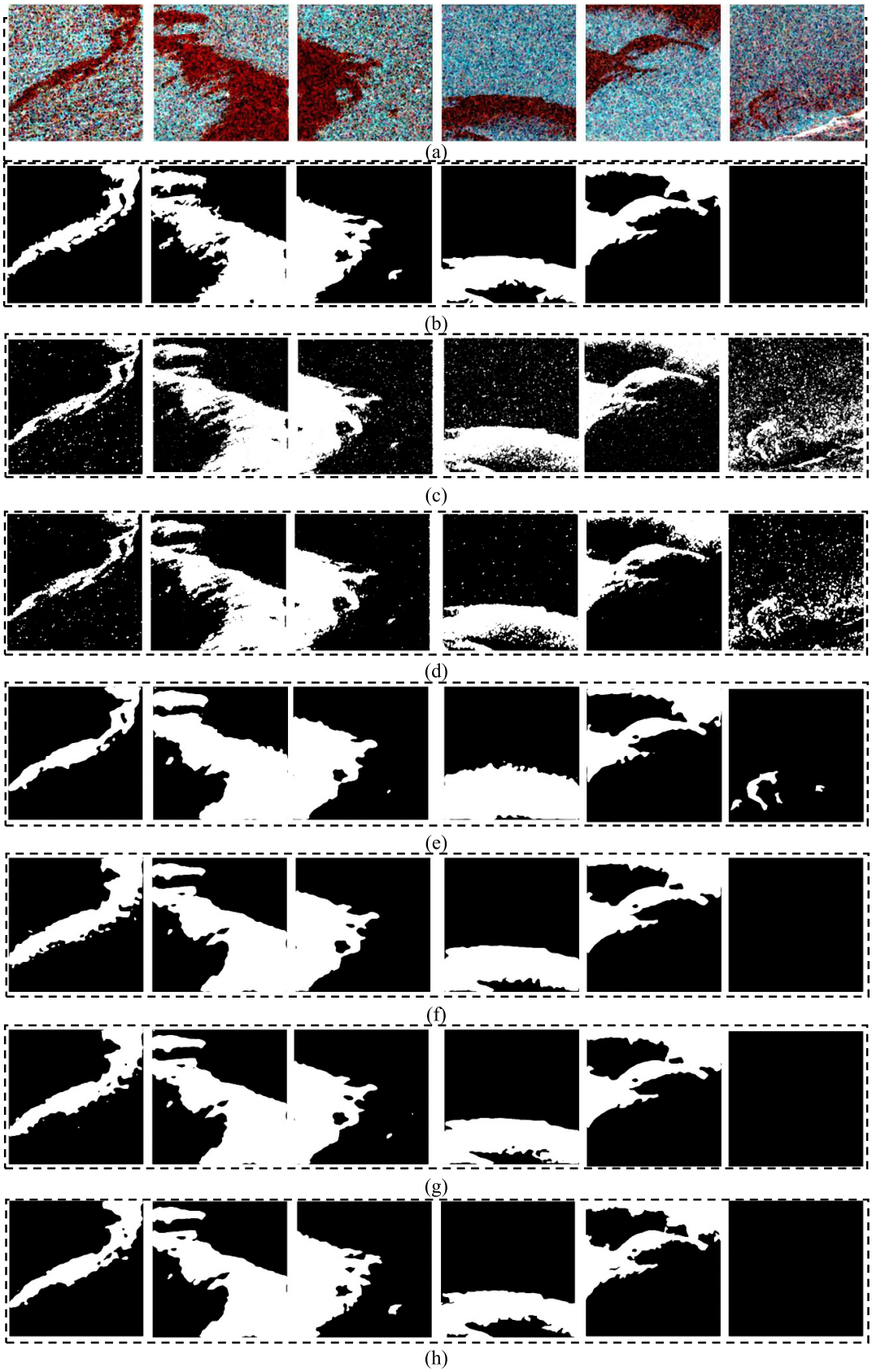


Fig. 9. (a) SAR images. (b) Ground truth of the SAR images. Examples of the detection results by using (c) SVM, (d) Wishart, (e) FCN, (f) SegNet, (g) U-Net, and (h) improved DeepLabv3+ model.

indicating that the model's learning ability is limited. When the backbone network is set to Resnet-101, the network reaches a stable state after the first few epochs, and the model achieves an mIoU of 95.55%. Furthermore, from Table IV, we can

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:07:39 UTC from IEEE Xplore. Restrictions apply.

TABLE V
PRECISION COMPARISON OF THE MODEL RESULTS

Model	Inference time per image (s)	OA (%)	F1 (%)	IoU_0 (%)	IoU_1 (%)	mIoU (%)
SVM	8.21	75.28	76.43	80.45	48.98	64.72
Wishart	4.37	82.15	74.83	83.57	61.19	72.38
FCN	0.86	87.98	85.48	96.38	80.15	88.27
SegNet	0.52	89.28	87.13	97.72	82.41	90.07
U-Net	0.26	96.77	**94.47**	**99.68**	90.29	94.99
Improved DeepLabv3+	**0.19**	**98.92**	94.38	99.52	**91.58**	**95.55**

see that the use of Resnet-101 as the backbone network of DeepLabv3+ can achieve a higher precision. Compared with other backbone networks, Resnet-101 has a deeper network structure and realizes shortcut connections between shallow and deep information of the image, which is beneficial to fully extract the useful features of the image.

Under different normalization methods, the BS is important for the model precision (see Fig. 8). When BS = 2, the OA, $F1$, and mIoU of the model using the GN layer are 98.92%, 94.38%, and 95.55%, respectively, which are approximately 1%–3% points higher than the results obtained when using the BN layer under the same parameter settings. The network with a GN layer also makes a lower demand on the memory of the computer than the network with a BN layer. This shows that, when the computer memory is limited, GN is an effective normalization method. Therefore, we believe that the DeepLabv3+ network based on a GN layer is more suitable for oil spill detection.

B. Comparison With Other Typical Models

In order to fully verify the performance of the model used in this study, the improved DeepLabv3+ model was compared with the traditional supervised classification methods (Wishart and SVM) and the classical semantic segmentation algorithms (FCN, SegNet, and U-Net).

The detection results are displayed in Fig. 9, and the quantitative assessment results are listed in Table V (the best results are in bold). It can be seen that the improved DeepLabv3+ model has the highest overall performance and has obvious advantages over the other models in terms of running time and classification accuracy.

From Fig. 9, it can be seen that the noise of the SAR images inevitably influences the traditional methods. The classification results of Wishart and SVM contain many spots, which seriously affects the accuracy of the classification. This indicates that the traditional methods only consider the shallow characteristics of the target while ignoring the contextual information. Therefore, the traditional methods cannot produce fine segmentation results. Furthermore, false positives for the oil spill instances are also apparent. For example, the original image in the right column is a nearshore and land area, with look-alike objects, such as low wind speed regions and land plants. These objects have polarimetric scattering traits that are similar to those of oil spills, leading to easy misjudgment by the traditional methods.

The semantic segmentation algorithms consider the contextual information, which can effectively reduce the impact of speckle noise in SAR images and improve the image classification quality. As shown in Table IV, the improved DeepLabv3+ model has the fastest inference time, which may be because the decoder of DeepLabv3+ uses a simple but effective module to recover the target boundary details, helping to further improve the speed and performance of the model. However, U-Net performs better than the improved DeepLabv3+ model in the $F1$ and IoU_0 values. When combined with the visual results, the improved DeepLabv3+ model can better capture the oil spill details and shows more robust performance in oil spill detection than FCN, SegNet, and U-Net. The above analyses imply that the improved DeepLabv3+ model is more suitable for oil spill detection.

V. CONCLUSION

Sea surface oil spill is one of the main threats to the marine ecological environment, so effective monitoring and early warning are needed. However, the detection of oil spills is a challenging task because of the complex nature of the surrounding environment. Inspired by the deep learning models that can automatically extract useful features from massive amounts of data, we used Sentinel-1 dual-polarimetric images as the data source while focusing on integrating multilayer features with a DCNN to detect oil spills, i.e., both the amplitude and phase information are utilized. Furthermore, to improve the feature discrimination capability of the proposed method, we also integrated Cloude polarimetric decomposition parameters into the model. Therefore, the network's learning process is based on the feature values of the pixels of the input image in the DCNN, with which in-depth mining of the polarimetric information of the PolSAR data is possible. The qualitative results showed that the speckle noise of SAR images inevitably influences the traditional classification methods, resulting in lower classification accuracies than the deep learning methods. Moreover, the experimental results described in this article proved that multilayer features (i.e., the polarimetric covariance matrix and H/A/alpha decomposition parameters) are a useful way to distinguish oil spills and look-alikes. Compared with the other classical DCNN semantic segmentation algorithms, the quantitative results showed that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:07:39 UTC from IEEE Xplore. Restrictions apply.

the improved DeepLabv3+ model, which takes ResNet-101 as the backbone network and GN as the normalization layer, generally achieved superior performance, in terms of both precision and inference time.

In our future work, we will focus on researching the oil spill detection capabilities of the model constructed under challenging conditions, such as lower winds and high winds. We will also attempt to improve the network to enhance further the model's efficiency for oil spill detection from PolSAR images.

REFERENCES

[1] D. C. Zacharias, K. F. O. Rezende, and A. Fornaro, "Offshore petroleum pollution compared numerically via algorithm tests and computation solutions," *Ocean Eng.*, vol. 151, pp. 191–198, Mar. 2018.

[2] P. Li, Q. Cai, W. Lin, B. Chen, and B. Zhang, "Offshore oil spill response practices and emerging challenges," *Mar. Pollut. Bull.*, vol. 110, no. 1, pp. 6–27, Sep. 2016.

[3] J. Chen, W. Zhang, Z. Wan, S. Li, T. Huang, and Y. Fei, "Oil spills from global tankers: Status review and future governance," *J. Cleaner Prod.*, vol. 227, pp. 20–32, Aug. 2019.

[4] (2020). *The International Tanker Owners Pollution Federation Limited Oil Tanker Spill Statistics*. [Online]. Available: http://www.itopf.com/knowledge-resources/data-statistics/statistics/

[5] W. Alpers, B. Holt, and K. Zeng, "Oil spill detection by imaging radars: Challenges and pitfalls," *Remote Sens. Environ.*, vol. 201, pp. 133–147, Nov. 2017.

[6] M. Krestenitis *et al.*, "Oil spill identification from satellite images using deep neural networks," *Remote Sens.*, vol. 11, no. 15, pp. 1762–1783, Aug. 2019.

[7] S. T. Yekeen and A. L. Balogun, "Advances in remote sensing technology, machine learning and deep learning for marine oil spill detection, prediction and vulnerability assessment," *Remote Sens.*, vol. 12, no. 20, pp. 3416–3446, Oct. 2020.

[8] H. Zheng, J. Zhang, Y. Zhang, A. Khenchaf, and Y. Wang, "Theoretical study on microwave scattering mechanisms of sea surfaces covered with and without oil film for incidence angle smaller than 30°," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 37–46, Jan. 2021.

[9] K. Topouzelis and A. Psyllos, "Oil spill feature selection and classification using decision tree forest on SAR image data," *ISPRS J. Photogramm. Remote Sens.*, vol. 68, pp. 135–143, Mar. 2012.

[10] A. Raeisi, G. Akbarizadeh, and A. Mahmoudi, "Combined method of an efficient cuckoo search algorithm and nonnegative matrix factorization of different Zernike moment features for discrimination between oil spills and lookalikes in SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 4193–4205, Jul. 2018.

[11] A.-B. Salberg, O. Rudjord, and A. H. S. Solberg, "Oil spill detection in hybrid-polarimetric SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 10, pp. 6521–6533, Oct. 2014.

[12] L. De Laurentiis *et al.*, "Deep learning for mineral and biogenic oil slick classification with airborne synthetic aperture radar data," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 10, pp. 8455–8469, Oct. 2020, doi: 10.1109/TGRS.2020.3034722.

[13] M. Migliaccio, A. Gambardella, and M. Tranfaglia, "SAR polarimetry to observe oil spills," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 2, pp. 506–511, Feb. 2007.

[14] F. Nunziata, A. Gambardella, and M. Migliaccio, "On the use of dual-polarized SAR data for oil spill observation," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2008, pp. 225–228.

[15] S. Skrunes, C. Brekke, and T. Eltoft, "An experimental study on oil spill characterization by multi-polarization SAR," in *Proc. 9th Eur. Conf. Synth. Aperture Radar*, Apr. 2012, pp. 139–142.

[16] R. Shirvany, M. Chabert, and J. Y. Tourneret, "Ship and oil-spill detection using the degree of polarization in linear and hybrid/compact dual-pol SAR," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 3, pp. 885–892, Jun. 2012.

[17] L. Xu, J. Li, and A. Brenning, "A comparative study of different classification techniques for marine oil spill identification using RADARSAT-1 imagery," *Remote Sens. Environ.*, vol. 141, pp. 14–23, Feb. 2014.

[18] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Convolutional neural networks for large-scale remote-sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 645–657, Feb. 2017.

[19] Y. Zhou, H. Wang, F. Xu, and Y.-Q. Jin, "Polarimetric SAR image classification using deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1935–1939, Dec. 2016.

[20] X. Yang *et al.*, "An attention-fused network for semantic segmentation of very-high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 238–262, Jul. 2021.

[21] X. Li *et al.*, "Deep-learning-based information mining from ocean remote-sensing imagery," *Nat. Sci. Rev.*, vol. 7, no. 10, pp. 1584–1605, Oct. 2020.

[22] G. Chen, Y. Li, G. Sun, and Y. Zhang, "Application of deep networks to oil spill detection using polarimetric synthetic aperture radar images," *Appl. Sci.*, vol. 7, no. 10, p. 968, Sep. 2017.

[23] K. Zeng and Y. X. Wang, "A deep convolutional neural network for oil spill detection from spaceborne SAR images," *Remote Sens.*, vol. 12, no. 6, pp. 1015–1037, Mar. 2020.

[24] F. M. Bianchi, M. M. Espeseth, and N. Borch, "Large-scale detection and categorization of oil spills from SAR images with deep learning," *Remote Sens.*, vol. 12, no. 14, pp. 2260–2286, Jul. 2020.

[25] H. Guo, D. N. Wu, and J. B. An, "Discrimination of oil slicks and lookalikes in polarimetric SAR images using CNN," *Sensors*, vol. 17, no. 8, pp. 1837–1856, Aug. 2017.

[26] J. Zhang *et al.*, "Oil spill detection in quad-polarimetric SAR images using an advanced convolutional neural network based on SuperPixel model," *Remote Sens.*, vol. 12, no. 6, pp. 944–969, Mar. 2020.

[27] (2012). *ESA's Radar Observatory Mission for GMES Operational Services*. [Online]. Available: https://sentinel.esa.int/documents/247904/349449/S1_SP-1322_1.pdfESA

[28] S. K. Chaturvedi, S. Banerjee, and S. Lele, "An assessment of oil spill detection using sentinel 1 SAR-C images," *J. Ocean Eng. Sci.*, vol. 5, no. 2, pp. 116–135, Jun. 2020.

[29] I. A. El-Magd *et al.*, "The potentiality of operational mapping of oil pollution in the mediterranean sea near the entrance of the Suez canal using sentinel-1 SAR data," *Remote Sens.*, vol. 12, no. 8, pp. 1352–1364, Apr. 2020.

[30] M. Konik and K. Bradtke, "Object-oriented approach to oil spill detection using ENVISAT ASAR images," *ISPRS J. Photogramm. Remote Sens.*, vol. 118, pp. 37–52, Aug. 2016.

[31] V. Kudryavtsev, "On effect of wave breaking on short wind waves," *Geophys. Res. Lett.*, vol. 31, no. 20, pp. 1–5, 2004.

[32] V. Kudryavtsev, "A semiempirical model of the normalized radar cross-section of the sea surface 1. Background model," *J. Geophys. Res.*, vol. 108, no. 3, p. 2, Jan. 2003.

[33] S. Skrunes, C. Brekke, C. E. Jones, and B. Holt, "A multisensor comparison of experimental oil spills in polarimetric SAR for high wind conditions," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 11, pp. 4948–4961, Nov. 2016.

[34] S. Singha, R. Ressel, D. Velotto, and S. Lehner, "A combination of traditional and polarimetric features for oil spill detection using TerraSAR-X," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 11, pp. 4979–4990, Nov. 2016.

[35] F. Girard-Ardhuin, G. Mercier, F. Collard, and R. Garello, "Operational oil-slick characterization by SAR imagery and synergistic data," *IEEE J. Ocean. Eng.*, vol. 30, no. 3, pp. 487–495, Jul. 2005.

[36] C. Brekke and A. H. S. Solberg, "Oil spill detection by satellite remote sensing," *Remote Sens. Environ.*, vol. 95, no. 1, pp. 1–13, Mar. 2005.

[37] G. Li *et al.*, "Marine oil slick detection using improved polarimetric feature parameters based on polarimetric synthetic aperture radar data," *Remote Sens.*, vol. 13, no. 9, pp. 1607–1629, Apr. 2021.

[38] M. M. Espeseth, C. Brekke, C. E. Jones, B. Holt, and A. Freeman, "The impact of system noise in polarimetric SAR imagery on oil spill observations," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4194–4214, Jun. 2020.

[39] M. W. Hansen, V. Kudryavtsev, B. Chapron, C. Brekke, and J. A. Johannessen, "Wave breaking in slicks: Impacts on C-band quad-polarized SAR measurements," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 11, pp. 4929–4940, Nov. 2016.

[40] M. Paquay, J. C. Iriarte, I. Ederra, R. Gonzalo, and P. D. Maagt, "Thin AMC structure for radar cross-section reduction," *IEEE Trans. Antennas Propag.*, vol. 55, no. 12, pp. 3630–3638, Dec. 2007.

[41] S. R. Cloude and E. Pottier, "An entropy based classification scheme for land applications of polarimetric SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 35, no. 1, pp. 68–78, Jan. 1997.

[42] S. Singha, A. M. Johansson, and A. P. Doulgeris, "Robustness of SAR sea ice type classification across incidence angles and seasons at L-band," *IEEE Trans. Geosci. Remote Sens.*, early access, Nov. 16, 2020, doi: 10.1109/TGRS.2020.3035029.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:07:39 UTC from IEEE Xplore. Restrictions apply.

[43] C. Lopez-Martinez and E. Pottier, "Statistical assessment of eigenvector-based target decomposition theorems in radar polarimetry," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2004, pp. 192–195.
[44] J.-S. Lee and E. Pottier, "H/A/α polarimetric decomposition theorem," in *Polarimetric Radar Imaging: From Basics to Applications*. Boca Raton, FL, USA: CRC Press, 2009, pp. 229–262.
[45] J. S. Lee, T. L. Ainsworth, J. P. Kelly, and C. Lopez-Martinez, "Evaluation and bias removal of multilook effect on entropy/alpha/anisotropy in polarimetric SAR decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 10, pp. 3039–3052, Oct. 2008.
[46] H. Yu *et al.*, "Methods and datasets on semantic segmentation: A review," *Neurocomputing*, vol. 304, pp. 82–103, Aug. 2018.
[47] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. ECCV*, Oct. 2018, pp. 833–851.
[48] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.
[49] M. Everingham, L. Van Gool, C. K. I. Williams. (2012). *The PASCAL Visual Object Classes Challenge 2012 (VOC2012) Results*. [Online]. Available: http://www.pascalnetwork.org/challenges/VOC/voc2012/workshop/index.html
[50] M. Cordts *et al.*, "The cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3213–3223.
[51] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
[52] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.
[53] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Sep. 2015.
[54] Y. Wu and K. He, "Group normalization," 2018, *arXiv:1803.08494*.
[55] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," 2015, *arXiv:1502.03167*.
[56] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1800–1807.
[57] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.
[58] M. Sandler *et al.*, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. 31st IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 4510–4520.
[59] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 3431–3440.
[60] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.
[61] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Oct. 2015, pp. 234–241.
[62] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 318–327.
[63] C. Tantithamthavorn, S. McIntosh, A. E. Hassan, and K. Matsumoto, "An empirical comparison of model validation techniques for defect prediction models," *IEEE Trans. Softw. Eng.*, vol. 43, no. 1, pp. 1–18, Jan. 2017.



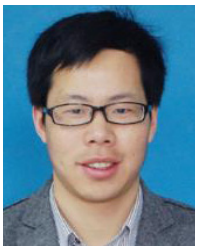
Xiaoshuang Ma (Member, IEEE) received the Ph.D. degree in cartography and geographic information engineering from Wuhan University, Wuhan, China, in 2016.

He is currently an Associate Professor with the School of Resources and Environmental Engineering, Anhui University, Hefei, China. His research interest includes synthetic aperture radar (SAR) image processing and interpretation.



Jiangong Xu received the B.S. degree in human geography and urban and rural planning from Hefei University, Hefei, China, in 2019, where he is currently pursuing the master's degree.

His main research is polarimetric synthetic aperture radar (PolSAR) image processing for sea oil spill intelligent identification.



Penghai Wu (Member, IEEE) received the M.S. degree in surveying and mapping engineering and the Ph.D. degree in cartography and geographical information engineering from Wuhan University, Wuhan, China, in 2011 and 2014, respectively.

He is currently an Associate Professor with the School of Resources and Environmental Engineering, Anhui University, Hefei, China. His research interests include land surface temperature reconstruction and fusion, deep learning, and regional ecoenvironmental change.



Peng Kong received the Ph.D. degree in microelectronics and solid state electronics from the Huazhong University of Science and Technology, Wuhan, China, in 2015.

He is currently an Engineer with the Institute of Spacecraft System Engineering, Beijing, China. His research interests include spacecraft system design, aerospace technology applications, remote sensing image analysis, and artificial intelligence.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:07:39 UTC from IEEE Xplore. Restrictions apply.